

Minimum Euclidean Distance

Time limit: 1.00 s

Memory limit: 512 MB

Given a set of points in the two-dimensional plane, your task is to find the minimum Euclidean distance between two distinct points.

The Euclidean distance of points (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

Input

The first input line has an integer n : the number of points.

After this, there are n lines that describe the points. Each line has two integers x and y . You may assume that each point is distinct.

Output

Print one integer: d^2 where d is the minimum Euclidean distance (this ensures that the result is an integer).

Constraints

- $2 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x, y \leq 10^9$

Example

Input	Output
4 2 1 4 4 1 2 6 3	2